EXPERIMENT 19

Word Sense Disambiguation Using the Lesk Algorithm

Aim

To identify the likely meaning of an ambiguous word using the Lesk algorithm.

Procedure

1. Install NLTK and download WordNet.
2. Import the Lesk function.
3. Prepare a sentence containing an ambiguous word.
4. Apply Lesk to the sentence and target word.
5. Print the selected synset and definition.

Program

```
# Install first: pip install nltk
import nltk
from nltk.wsd import lesk
from nltk.corpus import wordnet as wn

nltk.download("wordnet", quiet=True)
nltk.download("omw-1.4", quiet=True)

sentence = "He deposited money in the bank".split()
sense = lesk(sentence, "bank")
print("Selected sense:", sense)
if sense:
    print("Meaning:", sense.definition())
```

Output

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```
Selected sense: a WordNet synset for 'bank'
Meaning: definition of the selected synset
(The exact sense depends on the installed WordNet data and Lesk
    implementation.)
```

Result

Thus, the Lesk algorithm was used to select a likely sense of the ambiguous word from context.